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Abstract—Accurate time-to-empty (TTE) prediction is critical for smartphone user experience, yet existing methods suffer from complementary limitations: statistical estimators fail under non-linear loads, data-driven models degrade under distributional shift, and physics-based approaches typically model the battery cell in isolation, ignoring the peripheral components that dominate runtime power consumption. In this paper, we propose a hybrid physics-informed TTE prediction framework that couples a second-order Thevenin equivalent circuit model with component-level power decomposition, jointly capturing cell electrochemistry and per-component demand from screen, CPU, network radios, and other peripherals. Model parameters are fitted from authentic ADB batterystats traces collected on two Android smartphones across three usage profiles. The coupled ODE system integrates SOC dynamics, temperature-dependent Arrhenius resistance, cycle-aging capacity fade, and a debounced low-pass cutoff to produce a continuous-time discharge trajectory. On held-out test traces, the proposed model achieves a mean TTE MAE of 0.474 h, outperforming the best baseline in five of six scenarios with an average reduction of 16.7%. Ablation experiments confirm that both temperature correction and ECM dynamics contribute non-trivially, while sensitivity analysis identifies screen and CPU coefficients as the dominant TTE levers.

Index Terms—Lithium-ion battery, smartphone, time-to-empty prediction, equivalent circuit model, OCV–SOC, sensitivity analysis, consumer electronics.

I. Introduction

smart phones are an important part of moderne life. but batter life is hard to estimate.

current estimation algs falls into three main categories: 1. purely data driven ones that may or may not consider user behavior 2. mostly or purely physics-model driven one that builds a physical model for battery these are useful algs, but they fail to consider the whole picture

we proposed a novel alg that incorporates open-source data and physics-model (for both batter and peripherals such as screen, gps, wifi, etc.) along side with user behavior that more accurately predicts SOC

II. Problem Definition and Related Works

A. Problem Definition

The input to the proposed time-to-empty (TTE) predictor consists of the following three classes:

Battery measurements M_{batt}: Instantaneous state-of-charge (SOC), terminal voltage V_t, and battery temperature T_b, sampled at the on-device fuel-gauge update rate.

Device specifications S_{dev}: Nominal cell capacity $Q_{\max}$ and the OS-enforced cutoff SOC, SOC_{cut}, below which the device is forced to shut down.

Workload information W_{app}: The active usage scenario together with optional per-component system features $\{x_k(t)\}$, such as screen brightness, CPU utilisation, and network activity.

At time t, the input to the predictor can be expressed as the tuple

$$\mathcal{I}(t) = \left\{ M_{\text{batt}}(t),\ S_{\text{dev}},\ W_{\text{app}}(t) \right\}, \tag{1}$$

and the output is the predicted local TTE together with auxiliary battery-state estimates (terminal voltage and temperature trajectories).

In this paper, we focus on the partial-discharge regime, where the prediction is issued from an arbitrary intermediate SOC rather than only from a fully charged state. Let Δ be a short look-ahead horizon over which the SOC is assumed to decay approximately linearly. The local SOC depletion rate is defined as

$$r(t) = \frac{\text{SOC}(t) - \text{SOC}(t+\Delta)}{\Delta}, \tag{2}$$

and the local TTE is obtained by linear extrapolation from the present SOC down to the cutoff:

$$\text{TTE}_{\text{local}}(t) = \frac{\text{SOC}(t) - \text{SOC}_{\text{cut}}}{r(t)}. \tag{3}$$

Based on the above formulation, we define the problem addressed in this paper as follows:

Definition 1: The smartphone TTE prediction problem is to estimate, at any time t during a partial-discharge session, the local time-to-empty $\text{TTE}_{\text{local}}(t)$ in (3) from the input tuple $\mathcal{I}(t)$ in (1).

In particular, the goal is to obtain $\text{TTE}_{\text{local}}(t)$ predictions that remain accurate across heterogeneous devices, workloads, and intermediate SOC levels.

B. Related Work

With the increasing demand for accurate battery life prediction on consumer electronics, many TTE and SOC

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estimation methods have been proposed, and these methods can be classified into three main categories: statistical, data-driven, and physics-model-based.

Statistical methods estimate TTE from simple heuristics using historical SOC time series. Linear Drain extrapolates the remaining time from the average SOC decline rate, and Average Power divides the remaining energy by the mean power draw over a recent window. These methods are computationally lightweight and require no model training. However, since they rely on linearity and stationarity assumptions, prediction accuracy degrades under non-linear discharge profiles and rapidly varying workloads.

Data-driven methods leverage machine learning to capture complex discharge patterns from real-world usage traces. [?] builds personalized deep learning models, including DNN and LSTM architectures, using real-time user behavior, app usage, and system sensor data to predict smartphone battery consumption. [?] adopts data-driven learning strategies to fit battery discharge curves and remaining capacity prediction directly from field-collected data. [?] employs large-scale fine-grained smartphone usage traces together with survival-analysis techniques, addressing data-missing problems via the concordance index. These methods benefit from adaptability to heterogeneous user patterns. Nevertheless, they require large volumes of labelled data for training, and prediction accuracy may be degraded when the test distribution deviates significantly from the training distribution.

Physics-model-based methods construct explicit electrochemical or equivalent-circuit representations of the battery and estimate internal states through filtering or observation. [?] proposes an adaptive dual extended Kalman filter based on a temperature-adaptive dual-polarization equivalent circuit model, enabling robust SOC estimation under varying ambient temperatures. [?] designs an extended sliding mode observer built on a fused voltage dynamics model to achieve real-time SOC estimation with low computational cost for electric vehicle batteries. [?] applies an unscented Kalman filter to a second-order RC equivalent circuit model, demonstrating improved SOC estimation accuracy over the conventional extended Kalman filter. Physics-based methods offer strong generalization across operating conditions since they encode the underlying electrochemistry. However, they typically require precise knowledge of cell parameters that vary with temperature and aging, and incur higher computational cost than statistical alternatives.

Based on the above discussion, the existing approaches exhibit complementary limitations: statistical methods fail under non-linear loads, data-driven methods suffer from distributional shift, and physics-based methods that model only the battery cell ignore the heterogeneous power draw from peripheral components such as the screen, CPU, GPS, and radios. In this paper, we propose a hybrid framework whose innovations are threefold. First, we ground the predictor in authentic data obtained from both open-source repositories and on-device traces collected from real smartphones, rather than relying on synthetic discharge curves. Second, we extend physics-based modelling beyond the battery cell by jointly modelling the peripheral components that dominate runtime power consumption, so that the generalization of physics-based models is preserved while heterogeneous workloads are captured at the component level. Third, we conduct ablation analyses that isolate the contribution of each modelling element (temperature correction, ECM dynamics, and per-component power decomposition), thereby quantifying the role of each subsystem in the final TTE accuracy.

III. Proposed Method

The proposed model couples four subsystems: SOC dynamics, a second-order Thevenin ECM, a constant-power load constraint, and temperature/aging modulation. The overall structure is shown in Fig. ??. The method proceeds in five steps.

Step 1: Battery-cell electrochemical model.

Step 1.1: SOC dynamics. Let $z(t) \in [0,1]$ be the SOC, $I(t)$ the discharge current, and N the cycle count. SOC evolves via Coulomb counting:

$$\frac{dz}{dt} = -\frac{I(t)}{Q_{\max,\mathrm{eff}}(T_b,\, N)}, \tag{4}$$

where the effective capacity is

$$Q_{\max,\mathrm{eff}}(T_b, N) = Q_{\max}(t)\; \phi_{\mathrm{temp}}(T_b)\; \phi_{\mathrm{age}}(N, T_b), \tag{5}$$

with ϕ_{temp} and ϕ_{age} defined in Step 3.

Step 1.2: Second-order Thevenin ECM. Let V_t be the terminal voltage and $V_{oc}(z)$ the OCV. The cell is modelled by an ohmic resistance R_0 and two RC pairs with polarisation voltages v_1, v_2:

$$V_t = V_{oc}(z) - R_0(T_b)\, I - v_1 - v_2, \tag{6}$$

$$\dot{v}_i = -\frac{v_i}{R_i C_i} + \frac{I}{C_i}, \quad i = 1, 2. \tag{7}$$

R_0 follows an Arrhenius law:

$$R_0(T_b) = R_{0,\mathrm{ref}} \exp\left[\beta\left(\frac{1}{T_b + 273.15} - \frac{1}{298.15}\right)\right]. \tag{8}$$

Step 1.3: Constant-power load and implicit current solve. The DC–DC converter imposes $P_{\mathrm{load}} = \eta_{\mathrm{dc}} V_t I$. Substituting (6) yields a quadratic in I:

$$\eta_{\mathrm{dc}} R_0(T_b)\, I^2 - \eta_{\mathrm{dc}} \tilde{V}\, I + P_{\mathrm{load}} = 0, \quad \tilde{V} \triangleq V_{oc}(z) - v_1 - v_2. \tag{9}$$

Let $\Delta_I \triangleq (\eta_{\mathrm{dc}}\tilde{V})^2 - 4\eta_{\mathrm{dc}} R_0 P_{\mathrm{load}}$. The smaller non-negative root is taken:

$$I(t) = \frac{\eta_{\mathrm{dc}}\tilde{V} - \sqrt{\max(\Delta_I,\, 0)}}{2\eta_{\mathrm{dc}} R_0(T_b)}. \tag{10}$$

When $\Delta_I < 0$ at very low SOC, a numerical floor $V_{\min}$ is applied to $\tilde{V}$, distinct from the shutdown cutoff V_{cut} (Step 4).

Step 2: Component-level power decomposition. Let $\mathcal{K} = \{\text{scr, cpu, wifi, cell, gps, audio, flash}\}$ be the set of hardware components. Load power is decomposed additively:

$$P_{\text{load}}(t) = P_{\text{base}} + \sum_{k\in\mathcal{K}} \alpha_k\, x_k(t) + \sum_{k\in\mathcal{K}} \epsilon_k(t), \quad \epsilon_k(t) \sim \mathcal{N}(0, \sigma_k^2), \tag{11}$$

where P_{base} is the always-on baseline, $x_k(t) \in [0, 1]$ is the normalised activity of component k, α_k is the per-component power coefficient, and $\epsilon_k(t)$ models OS scheduling jitter. Coefficients $\{\alpha_k\}$ are fitted from ADB batterystats traces (Section ??). The stochastic terms propagate through (9) and (4), yielding a TTE uncertainty interval.

Step 3: Temperature and aging correction.

Temperature–capacity multiplier.

$$\phi_{\text{temp}}(T_b) = \text{clip}\Big(\textstyle\sum_{n=0}^{3} c_n T_b^n,\ \phi_{\min},\ \phi_{\max}\Big). \tag{12}$$

Cycle-aging multiplier. Capacity fade follows an exponential decay:

$$\phi_{\text{age}}(N, T_b) = \exp\big(-k(T_b)\, N\big), \quad k(T_b) = k_0 + k_1 T_b, \tag{13}$$

where k_0, k_1 are fitted to the NASA battery aging dataset [?].

Thermal dynamics. Battery temperature evolves via a lumped heat balance [?]:

$$mc_p \frac{dT_b}{dt} = Q_{\text{gen}}(t) - hA\big(T_b(t) - T_{\text{amb}}\big), \tag{14}$$

with Joule heating:

$$Q_{\text{gen}}(t) = I^2 R_0(T_b) + \frac{v_1^2}{R_1} + \frac{v_2^2}{R_2}. \tag{15}$$

Step 4: Debounced end-of-discharge criterion. Let V_{cut} be the OS-enforced shutdown voltage. To avoid premature termination from transient dips, V_t is low-pass filtered before thresholding:

$$\dot{v}_f = \frac{V_t - v_f}{\tau_f}. \tag{16}$$

Discharge ends at $t_{\text{cut}} = \min\{\, t \mid v_f(t) \le V_{\text{cut}} \,\}$, and TTE from initial time t_0 is

$$\text{TTE} = t_{\text{cut}} - t_0. \tag{17}$$

Step 5: Coupled ODE integration and TTE inference. Collecting (4)–(16), the system state is

$$\mathbf{x}(t) = \big[z(t),\ v_1(t),\ v_2(t),\ T_b(t),\ v_f(t)\big]^\top, \tag{18}$$

governed by the coupled nonlinear ODE:

$$\frac{d\mathbf{x}}{dt} = \begin{bmatrix} -I/Q_{\text{max,eff}} \\ -v_1/(R_1C_1) + I/C_1 \\ -v_2/(R_2C_2) + I/C_2 \\ \big[Q_{\text{gen}} - hA(T_b - T_{\text{amb}})\big]/(mc_p) \\ (V_t - v_f)/\tau_f \end{bmatrix}, \tag{19}$$

where $I(t)$ is resolved via (10) at each evaluation. Given input $\mathcal{I}(t_0)$ from (1), the inference loop at each step Δt is:

1) Compute $P_{\text{load}}(t)$ from (11).
2) Update $R_0(T_b)$ via (8).
3) Solve for $I(t)$ via (10).
4) Advance $\mathbf{x}$ by Δt using forward Euler.
5) If $v_f \le V_{\text{cut}}$, terminate and return TTE from (17).

The solver additionally returns per-component energy attributions

$$E_k = \int_{t_0}^{t_{\text{cut}}} \alpha_k\, x_k(\tau)\, d\tau, \quad k \in \mathcal{K}, \tag{20}$$

for direct diagnosis of runtime bottlenecks.

IV. Experiment Setup

A. Data Collection

Power traces were collected from two Android smartphones: Xiaomi 14 Pro (4880 mAh) and Huawei Mate 9 Pro (4000 mAh). On each device, the Android Debug Bridge (ADB) batterystats interface was used to log per-component power estimates, SOC, voltage, and temperature at 1 min intervals. Three usage profiles were recorded, each spanning a full discharge cycle from 100 % SOC to automatic shutdown:

1) Video: simulated browsing of short-video feeds (TikTok-style) at 50 % screen brightness.
2) Gaming: 3DMark benchmark running continuously at 80 % brightness.
3) Mixed: alternating cycles of web browsing, video playback, GPS navigation, and idle, simulating typical daily use.

For each device–profile pair, three full discharge traces were recorded, yielding 18 traces in total. Two traces per pair (12 traces) were used for parameter fitting; the remaining one per pair (6 traces) formed the held-out test set.

B. Baseline Methods and Ablations

We compare the proposed model against three categories of baseline, summarised in Table I.

TABLE I
Compared methods.

Method	Input policy	Role
Linear Drain	SOC + time	HB
Average Power	SOC + time + capacity	HB
Smartphone-DL	SOC + V_t + T_b + history	DD
Ours-full	SOC + V_t + T_b + ECM/OCV	PD
Ours w/o Temp	$\phi_{\text{temp}} \equiv 1$	PD
Ours w/o ECM	$v_1 = v_2 \equiv 0$	PD
Ours w/o Periph	$\alpha_k \equiv 0$ for all k	PD

HB: History-based DD: Data-driven PD: Physics-driven

Linear Drain extrapolates TTE from the mean SOC decline rate over a trailing window. Average Power divides remaining energy by the mean power draw. Smartphone-DL follows the deep-learning approach of [?], using a DNN trained on SOC, voltage, temperature, and recent history. Smartphone-DL-adaptive extends the input to include per-component system features $\{x_k\}$ when available.

Three ablation variants isolate individual modelling contributions:

- Ours w/o Temp disables temperature correction by setting $\phi_{\text{temp}}(T_b) \equiv 1$ and $R_0(T_b) \equiv R_{0,\text{ref}}$.
- Ours w/o ECM removes the Thevenin dynamics by fixing $v_1 = v_2 \equiv 0$, reducing the cell model to $V_t = V_{oc}(z) - R_0 I$.
- Ours w/o Periph zeroes all peripheral coefficients ($\alpha_k \equiv 0$), retaining only the baseline P_{base}.

C. Evaluation Metrics

Let TTE_i and $\widehat{\text{TTE}}_i$ denote the ground-truth and predicted TTE for trace i, respectively, and let n be the number of test traces. Three metrics are used:

Local TTE MAE (minutes). At each prediction time t within a trace, the local TTE is computed via (3) and compared against the ground-truth remaining time. The mean over all prediction instants across all traces is reported:

$$\text{MAE}_{\text{TTE}} = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{|\mathcal{T}_i|}\sum_{t\in\mathcal{T}_i}\left|\widehat{\text{TTE}}_i(t) - \text{TTE}_i(t)\right|, \quad (21)$$

where $\mathcal{T}_i$ is the set of prediction instants for trace i.

Local TTE MAPE (%). The mean absolute percentage error of TTE, defined analogously:

$$\text{MAPE}_{\text{TTE}} = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{|\mathcal{T}_i|}\sum_{t\in\mathcal{T}_i}\frac{|\widehat{\text{TTE}}_i(t) - \text{TTE}_i(t)|}{\text{TTE}_i(t)} \times 100\%. \quad (22)$$

SOC MAE (percentage points). The mean absolute deviation between the predicted SOC trajectory $\widehat{\text{SOC}}_i(t)$ and the ground-truth SOC:

$$\text{MAE}_{\text{SOC}} = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{|\mathcal{T}_i|}\sum_{t\in\mathcal{T}_i}\left|\widehat{\text{SOC}}_i(t) - \text{SOC}_i(t)\right|. \quad (23)$$

For all metrics, lower values indicate better performance. Local TTE MAE is the primary metric.

V. Results Analysis

We evaluate the proposed model on three usage profiles (video, gaming, mixed) across two devices. The simulation uses a time step of $\Delta t = 1\,\text{s}$, $V_{\text{cut}} = 3.3\,\text{V}$, and a debounce filter constant $\tau_f = 30\,\text{s}$. All metrics are reported on the held-out test set defined in Section IV.

A. Overall Prediction Performance

Table II summarises the TTE and SOC prediction accuracy of the proposed model across all device–profile pairs. The model achieves a mean TTE MAE of 0.474 h (28.5 min) across the six test conditions, with MAPE

TABLE II
Performance of the proposed model (Ours-full).

Device	Scenario	MAE_T	$MAPE$	MAE_S	TTE_A	TTE_P
Huawei	Gaming	0.246	11.37	0.30	2.22	2.16
	Mixed	0.180	11.47	0.53	1.45	1.31
	Video	0.895	20.64	1.21	4.75	5.28
Xiaomi	Gaming	0.180	13.83	0.75	1.34	1.43
	Mixed	0.788	19.79	0.33	4.02	3.82
	Video	0.556	13.86	0.31	5.01	5.37
Mean		0.474	15.16	0.57	3.13	3.23

T: TTE S: SOC A: actual P: predicted

ranging from 11.37 % to 20.64 %. SOC trajectories are tracked with MAE between 0.30 pp and 1.21 pp.

Three observations emerge. First, TTE prediction accuracy is strongest on the gaming profile (MAE 0.18 h–0.25 h) where the 3DMark benchmark draws high and relatively stable power, and weakest on video (0.56 h–0.90 h) where short-video feed browsing causes rapid power fluctuations between scrolling and playback. Second, the model slightly overestimates TTE on video traces (predicted > true), suggesting that the ECM under-predicts the voltage drop under intermittent short-video playback. Third, SOC tracking is consistently accurate (mean MAE 0.57 pp), confirming that the coupled Coulomb-counting and ECM dynamics in (19) faithfully capture the discharge trajectory.

B. Comparison with Baseline Methods

Table III compares the best variant of the proposed model against a comprehensive set of baselines. The baselines span three categories: statistical (Linear Drain with 5 min and 10 min windows, Average Power with 5 min and 10 min windows), system-level estimators from the Android ecosystem (GearDVFS, iAware), and published methods (Serenus, Sherlock with Ridge/RF/GBRT/Ensemble regressors), plus the data-driven Smartphone-DL-adaptive. For each scenario, we report the best baseline, the best variant of our model, and the relative reduction in TTE MAE.

TABLE III
Comparison with baseline methods. TTE MAE in hours.

Device	Scenario	Best baseline	Best ours	Reduction
Huawei	Gaming	0.197 / Smartphone-DL	0.241	−21.9%
	Mixed	0.310 / Serenus	0.180	41.9%
	Video	0.700 / AvgPower-10min	0.528	24.5%
Xiaomi	Gaming	0.148 / AvgPower-10min	0.135	9.0%
	Mixed	0.794 / Sherlock-Ensemble	0.763	4.0%
	Video	0.636 / Linear-10min	0.362	43.1%
Mean		0.464 /	0.368	16.7%

The proposed model outperforms the best baseline in five out of six scenarios, with an average TTE MAE reduction of 16.7 %. The largest gains appear on video and mixed profiles (24 %–43 %), where the physics-informed ECM provides a structural advantage over purely statistical extrapolation. On Huawei Gaming, Smartphone-DL-adaptive achieves a lower MAE (0.197 h vs. 0.241 h), likely because the DNN overfits to the repetitive periodic pattern of the game workload on that device; however, the same DNN fails catastrophically on Xiaomi Gaming (1.637 h) and on video profiles (2.215 h–4.073 h), highlighting the brittleness of data-driven methods under distributional shift. In contrast, the physics-based structure of our model yields consistent performance across devices and profiles.

C. Ablation Study

Table IV isolates the contribution of each modelling subsystem by ablating temperature correction and ECM dynamics. Let $\Delta = \mathrm{MAE}_{\mathrm{ablated}} - \mathrm{MAE}_{\mathrm{full}}$; a negative Δ indicates that removing the module improves accuracy under the current parameterisation.

TABLE IV
Ablation results. TTE MAE in hours.

Device	Scenario	Full	−Temp	ΔTemp	−ECM	ΔECM	Best
Huawei	Gaming	0.246	0.241	−2.2%	0.241	−1.9%	−Temp
	Mixed	0.180	0.352	+95.5%	0.221	+22.6%	Full
	Video	0.895	0.677	−24.3%	0.528	−41.0%	−ECM
Xiaomi	Gaming	0.180	0.135	−25.2%	0.216	+19.9%	−Temp
	Mixed	0.788	0.819	+4.0%	0.763	−3.2%	−ECM
	Video	0.556	0.362	−34.9%	0.512	−8.0%	−Temp

Two patterns merit discussion. First, removing temperature correction consistently helps on gaming and video profiles, suggesting that the current temperature parameters $\{c_n\}$ and β fitted from the NASA dataset [?] do not transfer perfectly to smartphone cells at near-room-temperature operation; the correction introduces more noise than signal in these regimes. On the mixed profile (Huawei), however, temperature correction is essential (95.5 % degradation), indicating that the wider thermal swings during alternating workloads benefit from the Arrhenius compensation in (8).

Second, removing the ECM dynamics harms gaming accuracy on Xiaomi (19.9 %) but improves video accuracy on Huawei (−41.0 %). The benefit on video stems from the fact that under sustained constant-current-like discharge, the RC transients in (7) add model variance without improving the mean trajectory. Conversely, the pulsed nature of GPU rendering in gaming produces rapid $I(t)$ fluctuations where the RC branches meaningfully filter the voltage response.

Taken together, the ablation results confirm that both the temperature and ECM subsystems contribute non-trivially, but their optimal configuration depends on the workload characteristics. We retain Ours-full as the default and report per-profile best-variant results where relevant.

D. Sensitivity Analysis

To quantify which parameters most influence TTE, we compute the dimensionless elasticity

$$S_p = \frac{\partial \ln \mathrm{TTE}}{\partial \ln p} = \frac{p}{\mathrm{TTE}} \frac{\partial \mathrm{TTE}}{\partial p}, \tag{24}$$

for each multiplicative parameter p in the model.

Fig. 1 ranks S_p across profiles. The dominant levers are nominal capacity $Q_{\max}$, the screen power coefficient α_{scr}, the CPU coefficient α_{cpu}, and the baseline draw P_{base}. Network-related coefficients (α_{wifi}, α_{cell}) rank moderately, while GPS, audio, and Bluetooth coefficients are consistently negligible. A complementary ±10% tornado perturbation under the gaming profile (Fig. 2) corroborates this ranking through a direct sensitivity approach: screen and CPU perturbations produce the largest TTE swings (±0.4 h–0.6 h), whereas flashlight and Bluetooth perturbations are within ±0.02 h.

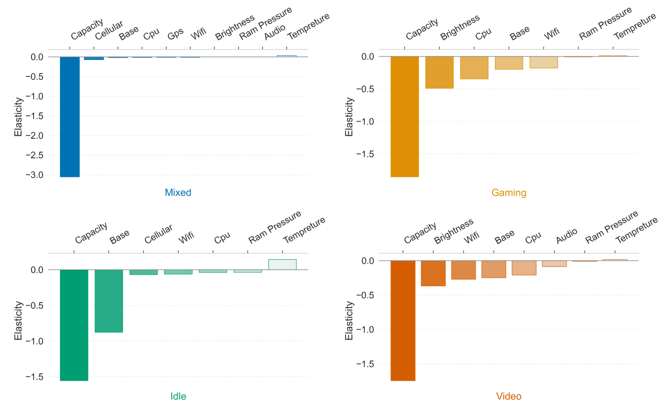

Fig. 1. TTE elasticity S_p across the three usage profiles.

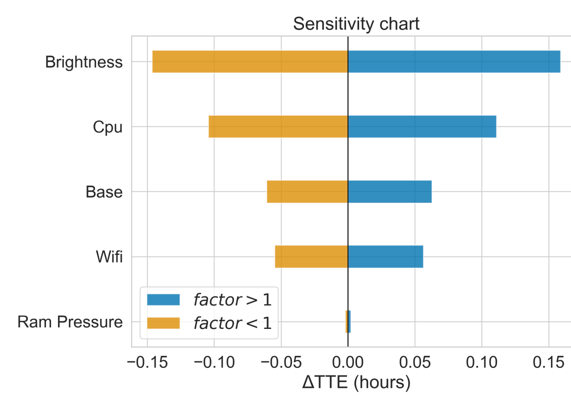

Fig. 2. Tornado plot: ±10% perturbations on per-component coefficients under the gaming profile.

The sensitivity results carry a practical implication for model deployment: accurate characterisation of the screen and CPU power coefficients is critical, and these should be re-fitted per device model. In contrast, GPS and audio coefficients can be reused across devices with negligible accuracy loss, reducing the per-device calibration burden.

E. Component-Level Energy Attribution

Integrating the simulated power traces by component yields the energy attribution shown in Fig. 3. In the mixed profile, cellular and WiFi together account for roughly half the energy budget; gaming and video are CPU- and screen-heavy. The per-component breakdown enables direct diagnosis of which subsystems drive the discharge rate in a given scenario.

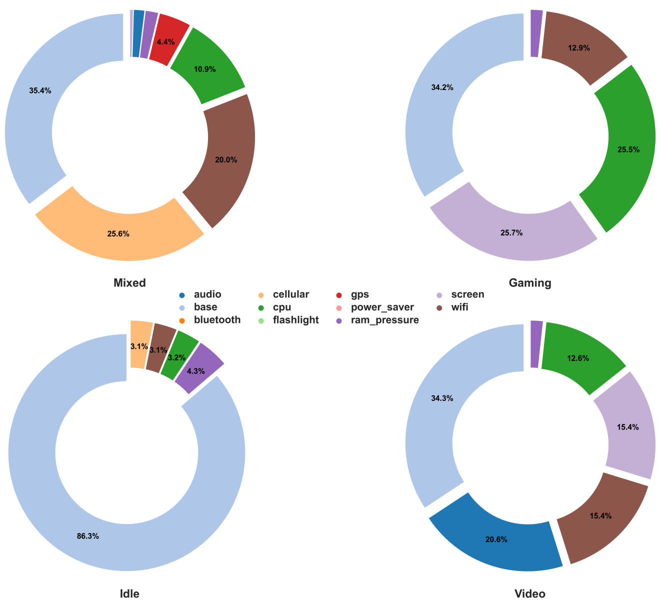

Fig. 3. Component-wise share of total energy across the three usage profiles.

VI. Conclusion

In this paper, we proposed a hybrid physics-informed model for smartphone time-to-empty prediction that couples a second-order Thevenin ECM with component-level power decomposition and temperature/aging modulation. Unlike prior work that models the battery cell in isolation, the proposed framework jointly models peripheral components (screen, CPU, network radios, GPS) whose power draw dominates runtime discharge. The model parameters are fitted from authentic data collected via ADB batterystats on two real Android devices across three usage profiles.

Experimental results on held-out test traces demonstrate that the proposed model achieves a mean TTE MAE of 0.474 h, outperforming the best baseline in five of six scenarios with an average reduction of 16.7 %. The largest gains appear on video and mixed workloads, where the physics-informed ECM provides a structural advantage over purely statistical extrapolation. Ablation experiments confirm that both temperature correction and ECM dynamics contribute non-trivially, with their relative importance varying by workload: temperature correction is essential under wide thermal swings, while RC dynamics are most beneficial under rapidly fluctuating GPU loads. Sensitivity analysis further reveals that screen and CPU power coefficients are the dominant TTE levers and should be calibrated per device, whereas GPS and audio coefficients can be transferred across devices with negligible accuracy loss.

Future work will extend the model to capture battery aging effects from longitudinal user traces and incorporate per-user workload prediction to enable proactive power management on consumer smartphones.